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Electrically tunable physical reservoir computing on two-terminal devices

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This work presents an electrically tunable physical reservoir computing (RC) system using two-terminal Pt/HfO_2/TiN memristors. The device's intrinsic relaxation time is effectively increased by over two orders of magnitude by modulating the base voltage between data pulses, enabling robust temporal data encoding for pulse intervals ranging from 0.14 to 100 ms. Enhanced separability is experimentally demonstrated through the classification of the Neuromorphic-MNIST (N-MNIST) dataset, surpassing the performance of conventional RCs without base voltage modulation. Frequency-domain analysis using multiple superimposed oscillator inputs shows controllable low-pass filtering characteristics. At the same time, the chaotic time-series prediction of the Mackey–Glass dataset achieves a normalized root mean square error of 0.27 without input data modulation and teaching signal for 200 timesteps. These results illustrate a feasible pathway toward tunable RC systems that achieve high performance while keeping the simplicity of a two-terminal device architecture.

Reservoir computing (RC) is a robust method that efficiently processes temporal data. It utilizes recurrent and complex connections between the reservoir nodes to encode temporal data, and is therefore also known as an echo state machine[1,2]. Complex dynamics of the reservoir enable a simple readout process without the energy-heavy training process of the reservoir itself[3]. Although it was started in computer science, RC has been implemented in various physical systems for energy efficiency[4–6]. Starting from the work of Du et al, short-term memories were widely used to replace complex reservoir dynamics with fading memory[5]. Compared to those based on optical systems[6] or field-programmable gate arrays[7], electrical devices such as memristors[8,9] and semiconductors[10] showed reduced complexity, enabling various tasks, such as signal processing[11,12], image classification[13,14] and medical analysis[15], to be feasibly conducted with devices' intrinsic relaxation.

These physical RC works typically aim to achieve two main goals: separability and tunability. Separability refers to the system's ability to distinguish different temporal (or spatial) inputs, which measures how well the system encodes the input data into a more complex nonlinear space. The most common approach is to use an input mask, which applies fixed or random transformations to the input signal, such as temporal shifts or amplitude changes, to increase diversity and separability in the reservoir's internal states[16,17]. In addition, complex device engineering is often employed to enhance separability. For instance, additional electrical (or optical) stimulus is used to modify the characteristics of the channel in three-terminal devices, used as the reservoir[14,18,19]. Other approaches include control of the switching dynamics through materials engineering, complex array formation[15,20,21], and using additional circuit elements[22].

On the other hand, tunability refers to the system's ability to adapt to various data types and timescales. The system must operate effectively across various environments, especially when timescales vary[23]. In physical RC, however, tunability is constrained by the intrinsic timescale of the devices. Although tunability is an essential metric for scalable RC, it is often treated as a secondary means to achieve separability, the most crucial metric for a physical RC. In this regard, several approaches have been employed to enhance tunability, aiming for greater separability. Jang. et al. utilized an additional resistor-capacitor circuit delay to extend the timescale beyond the device limit[22]. Liu et al. used additional opto-electrical stimuli in three-terminal In_2Se_3 thin film transistor devices to achieve multi-scale RC, where the relaxation timescales were dynamically tuned via light intensity and back-gate voltage to control timescale response in dynamic RC[24]. Yoo et al. adjusted the magnesium concentration in the entropy-stabilized oxide memristor (($Mg_xCo_{(1-x)/4}Ni_{(1-x)/4}Cu_{(1-x)/4}Zn_{(1-x)/4}$)O) on top of the epitaxial $YBa_2Cu_3O_{7-x}$ bottom electrode, to tune the local structural disorder and modulate the memristor's switching dynamics. It enabled control of the

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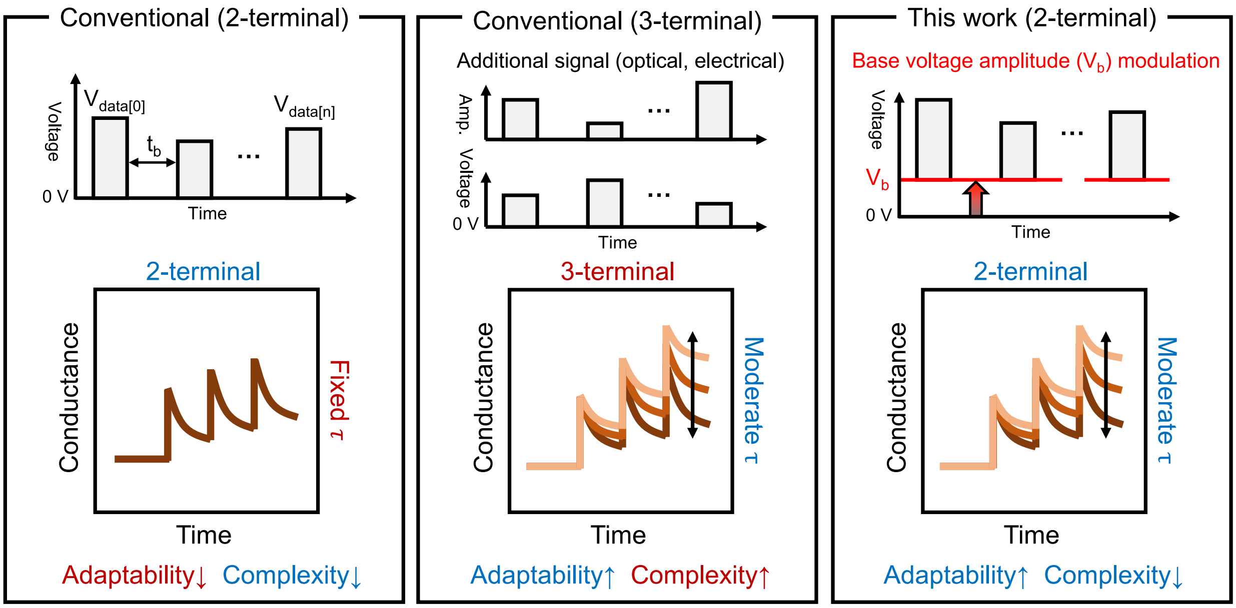

Fig. 1 | Schematic comparing conventional RC approaches. Conventional methods use two-terminal devices with encoded data pulses (left panel) or three-terminal devices with additional optoelectronic control (middle panel). Each method takes a trade-off between system adaptability and complexity. In contrast, the proposed method on the right panel modulates V_b in a two-terminal PHT device for high tunability and reduced complexity.

internal time constant[25]. However, the previous works either lack flexibility in structure or require extensive materials engineering and complex device structures.

Figure 1 depicts the tunability of short-term memory and its timescale, where the left and middle panels depict two conventional approaches in a physical reservoir. The first conventional approach, based on two-terminal devices (left panel), relies solely on the volatile nature of the material. While this method offers a simple structure, it lacks tunability[26,27]. The second conventional approach, which employs three-terminal devices (middle panel), introduces an additional terminal and optoelectronic signals to enable tunability at the cost of increased complexity[19,21]. Conversely, this work modulates base voltage (V_b) amplitude of the input voltage signal pulses, as shown in the right panel, to modify the timescale, resulting in high tunability and reduced complexity.

This study proposes an electrically tunable physical RC system based on a two-terminal Pt/HfO_2/TiN (PHT) device[28], where the V_b amplitude is modulated between voltage pulses representing the inputs. An approximately three-order expansion in the device's timescale (tunability) was achieved without requiring complex device structure or additional stimuli. Therefore, enhanced separability was achieved by utilizing a range of V_b and the device's full dynamic range, even for data with an extended timescale. This work first discusses the electrical characteristics of PHT devices with varying V_b. Then, the Neuromorphic-MNIST (N-MNIST) dataset[29] was classified to evaluate the system's tunability. Finally, the tunability of the reservoir was quantitatively evaluated using the multiple superimposed oscillators (MSO) dataset, and its achieved separability was demonstrated in the Mackey-Glass (MG) dataset, highlighting its applicability in predicting chaotic time series.

Results

Electrical characteristics of the PHT

A unit memristor should exhibit rectifying properties, fading memory, and minimal variations for reliable reservoir operation in passive array architectures. The PHT memristor used in this study consists of a Pt top electrode/4 nm-thick HfO_2 layer/TiN bottom electrode with an electrode area of 80 μm^2. The HfO_2 layer was deposited via thermal atomic layer deposition (ALD), while the Pt and TiN layers were deposited through sputtering. Supplementary Fig. 1 shows a cross-sectional transmission electron microscopy image of the Pt/HfO_2/TiN stack. The methods section provides further details on the fabrication process. Figure 2a presents 600 consecutive I-V cycles of the PHT device, illustrating its rectifying switching characteristics and minimal cycle-to-cycle variation. The device exhibits gradual switching behavior during both set (switching from the high-resistance state (HRS) to the low-resistance state (LRS)) and reset (switching from the LRS to HRS) operations, enabling multilevel switching characteristics without requiring the application of a compliance current. It has been reported that the set and reset switching occur via the trapping and detrapping of electrons from and to the TiN bottom electrode.

At the same time, the high Schottky barrier at the Pt/HfO_2 interface causes rectification[28,30] due to the work function difference between the top Pt (~5.5 eV) and TiN (~4.5 eV) electrodes. Local electron trapping creates a conduction channel in the HfO_2 film, where trap-assisted tunneling conduction predominates. As a result, it enables electroforming-free behavior, unlike filamentary memristors that need local filaments to form.

Figure 2b shows the HRS and LRS current values, measured by 600 DC cycles with the sweep voltage ranging from −2.5 to 3 V and a read voltage of 1.85 V, indicating high endurance. Figure 2c displays the device-to-device variation measured across 36 memristors, showing consistent operation with minimal variation.

In addition to exhibiting stable DC characteristics essential for array reliability, the PHT device also displays two key dynamic behaviors crucial for reservoir operation under AC conditions: relaxation and potentiation. Figure 2d illustrates the temporal relaxation of conductance measured in the device, initially set to the high-conductance state (~0.25 μS), under $V_b = 0$ V, using 1 ms/1.85 V read pulses over a 20 ms period. The black dots represent measured data, while the red lines show the fitted results using an exponential function, yielding a characteristic relaxation time (τ) of 0.92 ms. This relaxation time can be further modulated by adjusting the V_b.

Complementary to this relaxation (or natural depression) behavior, Fig. 2e presents the potentiation behavior of the device, evaluated by applying various set voltages under $V_b = 0$ V. Each plot presents the average and standard deviation from five repeated measurements, where the write (set) and read pulses, represented by the inset figure, are repeatedly applied. High-amplitude set pulses (1.9–4.1 V) induce a temporary conduction channel in the HfO_2 layer, increasing conductance through a trap-assisted mechanism. In contrast, lower set voltages (<~1.85 V) result in negligible

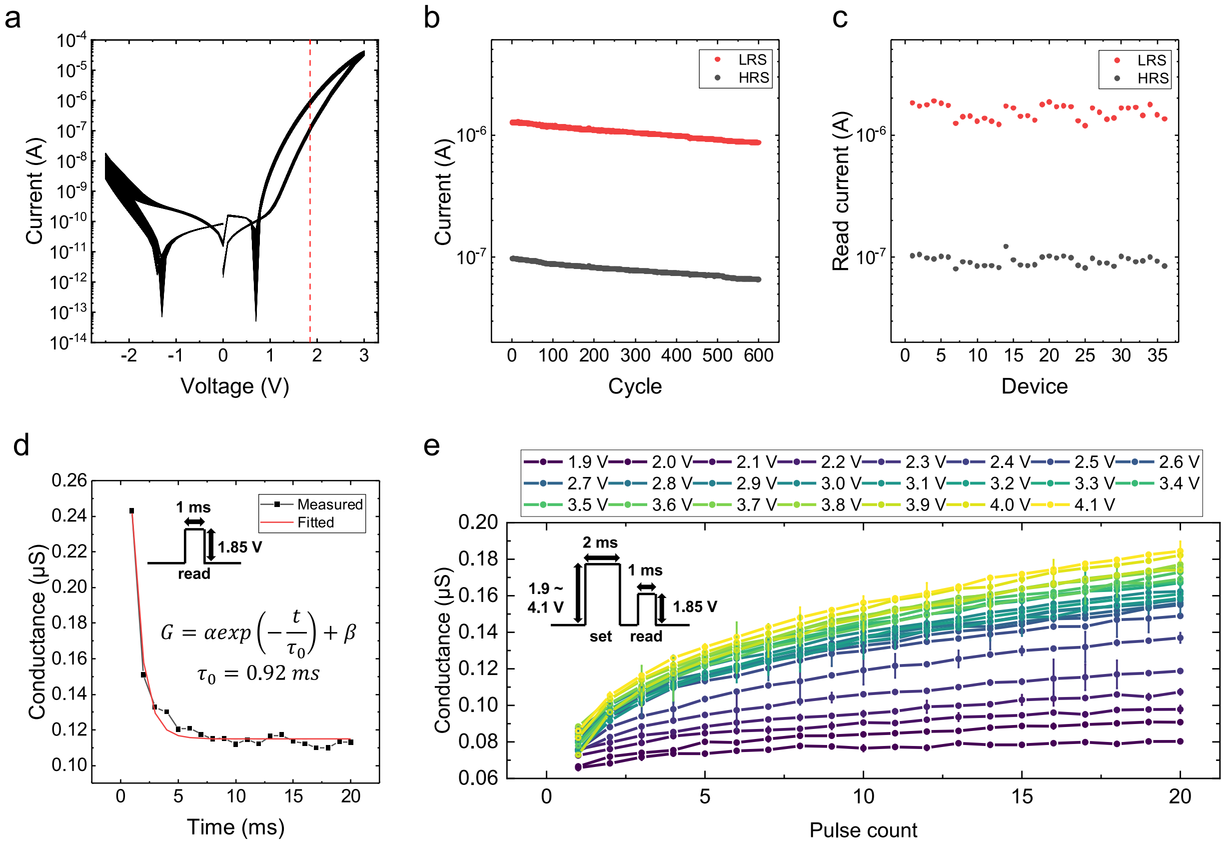

Fig. 2 | Electrical characteristics of the PHT memristor. a I–V characteristics of the PHT memristor, showing 600 consecutive I–V sweeps from −2.5 V to 3.0 V. **b** Read current at HRS and LRS (measured at 3.0 V set) over 600 DC cycles. Both states exhibit high endurance and minimal cycle-to-cycle variation over 600 cycles, with uniform current distributions for LRS (red dots) and HRS (black dots). **c** HRS and LRS current levels measured across 36 different PHT devices, demonstrating D2D uniformity. **d** Conductance relaxation behavior of the PHT memristor. Black dots represent the measurement data, while the red lines indicate the fitted modeling results. **e** Conductance response as a function of set pulse count. Pulses are applied at 1 ms intervals, and the results represent the mean and error bars from ten measurements for each condition.

changes in conductance. This potentiation behavior reflects short-term memory dynamics and, therefore, can be modulated by adjusting the V_b.

The relaxation and potentiation behaviors shown in Fig. 2d and e are fundamental mechanisms widely utilized in physical RC to encode temporal data. Temporal data is represented by voltage pulses that induce conductance increases through potentiation. Conversely, the absence of temporal input is captured through conductance relaxation, allowing the device to encode inactivity and intervals.

Figure 3a illustrates the concept of the proposed pulse scheme for tuning the reservoir's temporal dynamics by modulating the base voltage (V_b). The first and third panels from the top illustrate the conventional method, where 4-bit patterns "0110" and "1010" are encoded using a constant V_b of 0 V. In contrast, the second and fourth panels depict the proposed method, where the same inputs are encoded with a modulated V_b to dynamically adjust the relaxation behavior during the input pose duration (t_b). This approach enhances the system's tunability across a wide range of temporal scales. Before applying complex or irregular input patterns, the system's relaxation and potentiation response to uniform and straightforward temporal input sequences (e.g, "000…" or "111…") was evaluated under various t_b and V_b combinations.

First, Fig. 3b–e show the conductance relaxation in the high-conductance state (~ 0.25 μS), monitored through 1 ms/1.85 V read pulses applied up to 20 times. The device was initially set to the ~ 0.25 μS conductance state using a 2 ms/4.1 V pulse, followed by reading steps with the V_b and t_b modulations between the read pulses. For each t_b value of 0.14, 1.0, 10, and 100 ms, V_b was varied from 0 to 1.6 V. In all cases, the result indicates conductance decreases rapidly during the initial pulses, attributed to the detrapping of trapped electrons within the HfO_2 film. The conductance decrease becomes more substantial as the t_b increases and V_b decreases. While t_b is typically determined by the target application, such as input data frequency or timing, V_b remains a controllable parameter, enabling the fine-tuning of the relaxation time constant (τ) to match specific system requirements. The τ values corresponding to each t_b–V_b condition are determined via exponential fitting, following the same method used in Fig. 2d.

Figure 3f shows that the relaxation time constant increases with higher V_b, reaching a maximum τ of 123 ms at t_b=100 ms and V_b = 1.6 V. This corresponds to an increase over 100 times compared to the baseline relaxation timescale τ_0 = 0.92 ms at V_b=0 V (Fig. 2d), demonstrating that the proposed method effectively extends τ to accommodate longer input durations. Moreover, the minimum τ observed for each t_b increases with increasing t_b. This trend is attributed to the fact that most relaxation occurs within the intrinsic timescale τ_0, and for t_b >1 ms, this rapid initial relaxation is not captured due to the delayed timing of the first read pulse. This effect is evident in Fig. 3d and 3e, where most of the relaxation precedes the first read. Figure 3g presents the relative relaxation timescale $\tau/\tau_{Vb.0\ V}$, where $\tau_{Vb.0\ V}$ is the minimum τ for each t_b at V_b = 0 V. The relative timescale increases with V_b for all t_b, indicating that stronger base voltage application delays the relaxation behavior. At a given V_b, the relative timescale increases with increasing t_b up to 1 ms. Still, it decreases again when t_b increases to 10 and 100 ms, where the minimum τ is overestimated due to the readout timing.

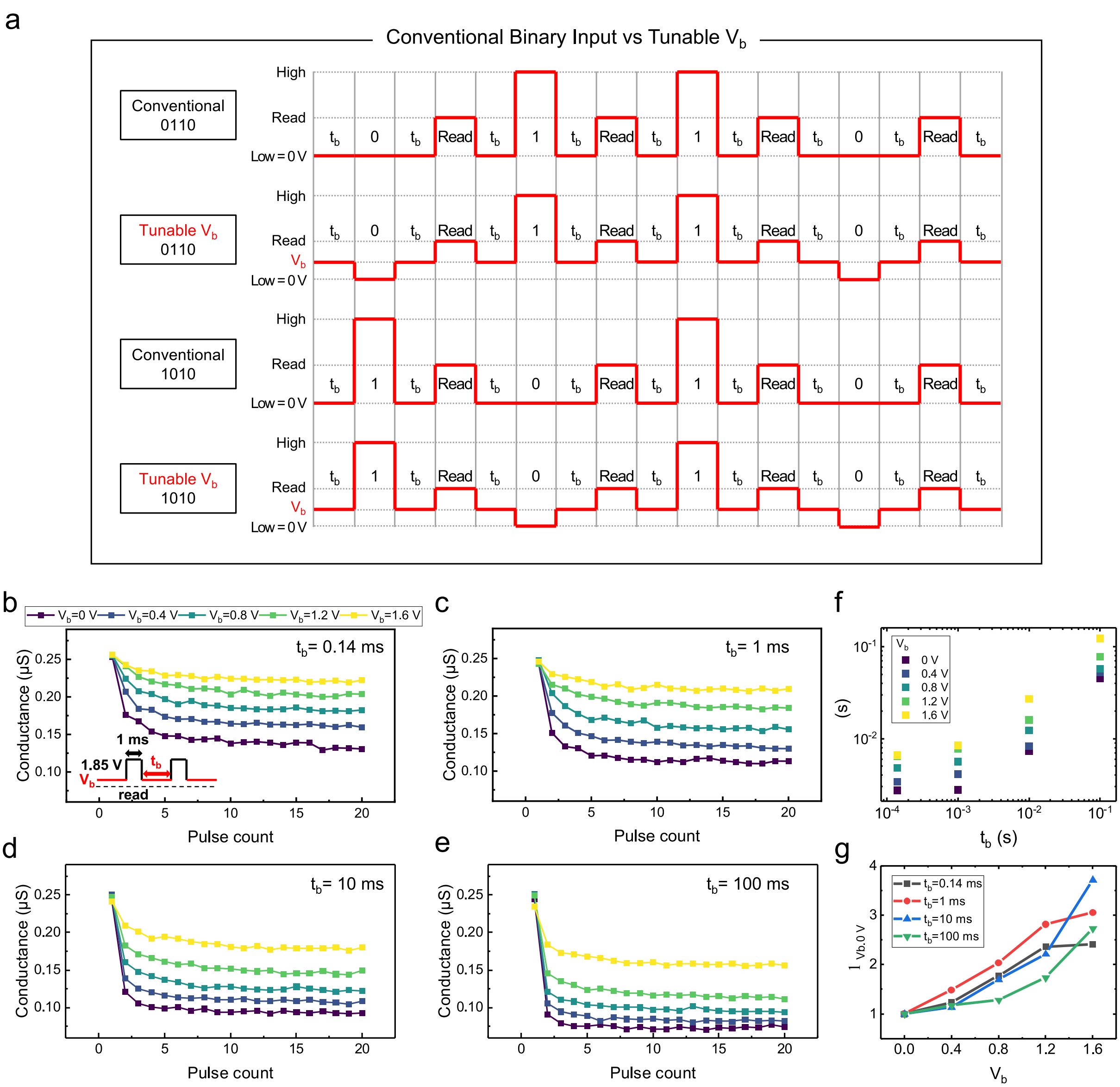

Fig. 3 | Relaxation of the PHT's short-term memory and its tunability. **a** Schematic comparison between conventional binary input and tunable V_b. In the conventional binary input, the voltage remains constant at 0 V during t_b, whereas in the tunable V_b scheme, the voltage is adjusted to V_b. **b**–**e** Measured LRS conductance relaxation under varying V_b from 0 to 1.8 V and t_b of 0.14, 1.0, 10, and 100 ms. Each result shows the average from five independent measurements. The conductance decreases more significantly with longer t_b and lower V_b, due to enhanced electron de-trapping. **f** Relaxation timescale range from various t_b. The minimum and maximum values are drawn from V_b of 0 and 1.6 V, respectively. **g** Relative relaxation timescale $\tau/\tau_{Vb,0\ V}$ as a function of V_b, where $\tau_{Vb,0\ V}$ refers to the minimum τ at V_b = 0 V for each t_b. The relative value increases as V_b increases, but the overestimation of $\tau_{Vb,0\ V}$ for t_b > 1 ms results in a decrease compared to shorter t_b.

This modulation of the relaxation time constant originates from the internal switching mechanism of the PHT device, as illustrated in Supplementary Fig. 2. When a positive bias is applied to the Pt electrode, electrons are trapped within the HfO_2 layer, increasing conductance through a trap-assisted mechanism[28,31]. Besides, the work function difference (~1 eV) between Pt and TiN leads to the formation of an internal field, which de-traps the trapped electrons in the HfO_2 layer (third panel) and thereby relaxes the LRS conductance. A positive V_b compensates for this internal field formation, inhibiting electron detrapping and effectively increasing τ. A further explanation is provided in Supplementary Note 1.

Then, the potentiation of the PHT with different τ was also attempted by varying V_b to for real-time RC applications. Given a dataset with an extended timescale, the intrinsic τ of the PHT device may be insufficient to retain the data history as conductance. From one to five binary potentiation pulses representing input data with different t_b values from 0.14 to 100 ms were used, and conductance was measured with different V_b. Each data point is obtained from independent measurements, where N set pulses (N = 1–5) are applied, followed immediately by a read pulse (see inset of Fig. 4a, second panel). Up to five consecutive pulses of 2 ms/3 V were applied to the PHT, and each panel shows the results from six t_b values. The condition of V_b of 0 V corresponds to the conventional RC on two-terminal devices. Under this condition, the device conductance increases with the increasing number of pulses, reaching ~0.16 μS after the five pulses, when t_b is 0.14 and 1 ms, comparable to the device's intrinsic τ (~0.92 ms). When an appropriate V_b (>~1.0 V) is applied, the device's conductance increase is strengthened, reaching ~0.2 μS after the five pulses. These properties allow the input pulse number to be easily encoded into the device conductance. However, when t_b becomes longer than 1 ms with V_b < ~1.0 V, the system fails to encode temporal data because the temporally increased conductance is relaxed to its initial value during the extended duration between pulses.

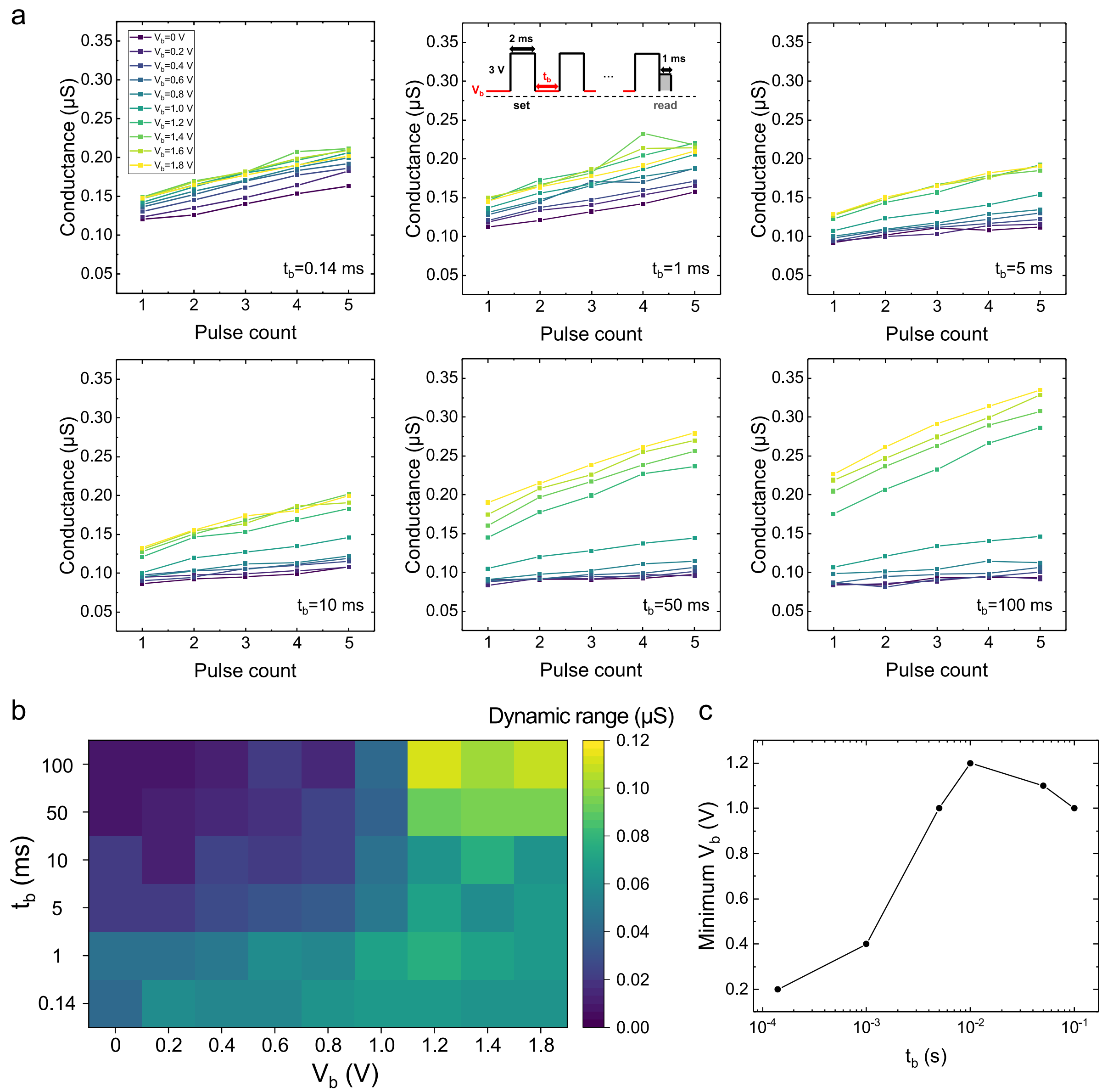

Fig. 4 | Potentiation of the PHT's short-term memory and its tunability. **a** Conductance response curves for five consecutive set pulses (3 V) recorded under different V_b and t_b conditions. The input pulses, with six intervals ranging from 0.14 to 100 ms, represent different data timescales. The conductance values were measured directly after the last potentiation pulse. Each panel shows the averaged results from five measurements. **b** Dynamic range as a function of t_b and V_b. The maximum value is approximately 0.12 μS. The dynamic range becomes smaller with longer t_b but increases with higher V_b. **c** Minimum V_b required to achieve a dynamic range greater than 0.04μS for each t_b. Longer t_b requires higher V_b for the device to encode consecutive pulses.

Specifically, the conductance remains at ~0.1 μS irrespective of the pulse counts, which is insufficient for the system to encode temporal information reliably.

In contrast, the proposed method maintained its dynamic range even for prolonged input pulses (t_b > 1 ms) by applying higher V_b levels, preventing conductance relaxation. The dynamic range means the difference between the conductance values after the first and fifth pulse applications. For example, the final conductance becomes ~0.2 μS when V_b > ~1.6 V, which can feasibly encode the temporal input into the device conductance. This demonstrates that V_b modulation enhances separability in pulse-induced potentiation by fully utilizing the device's dynamic range, extending its effects beyond the intrinsic timescale. However, excessive V_b (>~1.2 V) in longer t_b (>50 ms) resulted in potentiation of the device during base voltage application, significantly increasing conductance to approximately 0.3 μS. This behavior is due to the trapping of electrons in the HfO_2 layer for t_b over 50 ms, which is undesired. It was desired that the electron trapping (thus the potentiation) occurred only during the write pulse application, while the V_b during t_b was supposed only to suppress the electron loss. In this case, the conductance increase after the first pulse application becomes excessive, and the dynamic range does not increase despite the increased device burden. Supplementary Fig. 3 shows the potentiation characteristics among five different devices, exhibiting similar characteristics.

Figure 4b shows the dynamic range variation as a function of t_b and V_b, where the maximum value is ~0.12 μS, which is reasonably high for this work. The results indicate a smaller dynamic range for prolonged t_b, which can be increased by increasing V_b. Nonetheless, it cannot be arbitrarily increased for the reason mentioned above. Figure 4c presents the minimum V_b required for the PHT device to achieve a dynamic range > 0.04μS for the various t_b values. The threshold of 0.04 μS was selected based on the dynamic range observed at t_b = 0.14 ms, which is below the intrinsic time

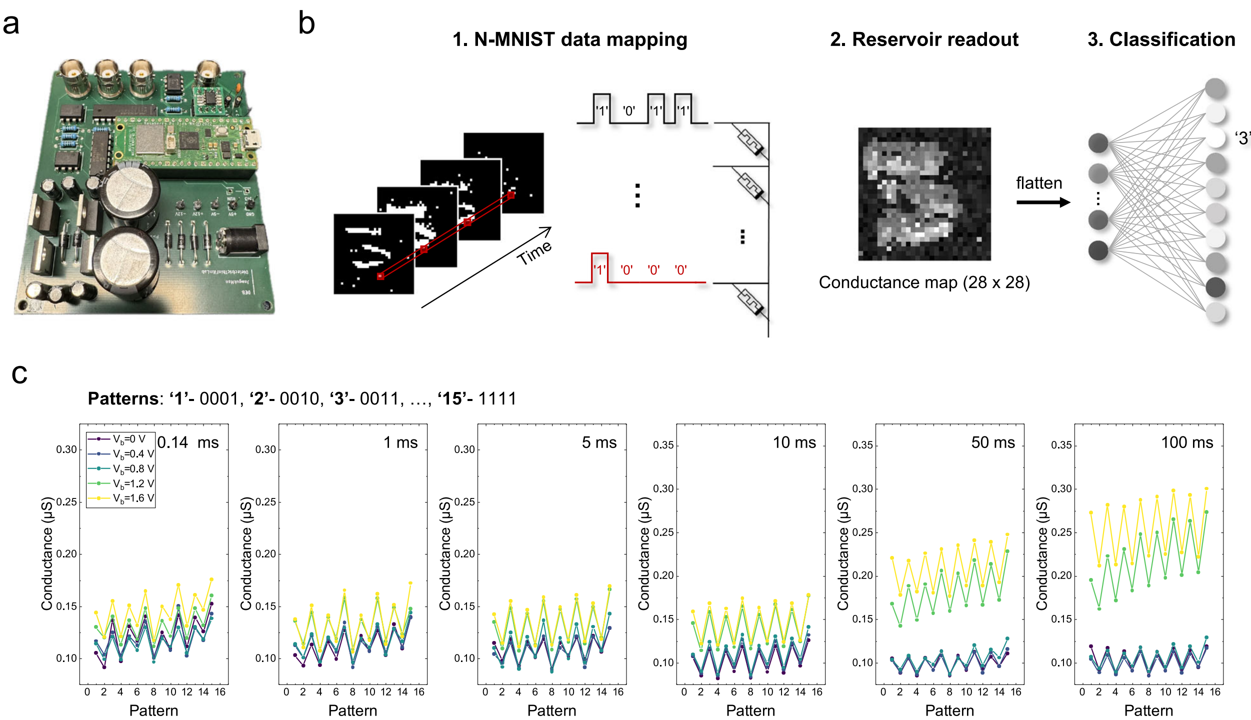

Fig. 5 | Experimental validation of tunability and separability in N-MNIST classification. **a** The experimental setup for N-MNIST classification. **b** Flow diagram outlining the classification process. Temporal pixel data on the left is converted into binary pulses that modulate the conductance states of individual memristors. The second step shows a 28 × 28 conductance map of serial readout of 784 PHTs. Then, conductance values are directed to a readout layer for classification. **c** Averaged conductance responses from 15 distinct 4-bit binary pulse patterns across various t_b in five measurements. It demonstrates that increased V_b improves separability even at prolonged pulse intervals, in contrast to conventional RC (V_b = 0 V).

constant τ. This value represents the baseline modulation range achievable purely from the device's intrinsic dynamics, without relying on enhanced voltage-driven separation. These results demonstrate that higher V_b is needed to attain the dynamic range necessary for scalable RC, which requires more complex engineering in conventional RC approaches.

Experimental validation on N-MNIST

Based on the relaxation and potentiation measurements, the tunability of RC with V_b modulation was experimentally validated using the custom-built measurement setup shown in Fig. 5a. The hardware consists of a digital-to-analog converter, a transimpedance amplifier, and an analog-to-digital converter, enabling flexible pulse readout (see the methods section and Supplementary Fig. 4 for details of this custom setup). The N-MNIST dataset, a neuromorphic adaptation of the standard MNIST dataset, was utilized to encode spatiotemporal event-based information to demonstrate the performance of the proposed physical RC[29]. Unlike the standard MNIST dataset[32], where digits from 0 to 9 are represented as static images, the N-MNIST dataset consists of event-based on-spike data of the digits. While previous works achieved comparable classification accuracy and lower energy consumption than software deep neural networks using a single-layer readout, the use of static image datasets fails to capture the complexities of real-world scenarios, where data is generated asynchronously and across diverse timescales[23,33]. The N-MNIST dataset offers a more realistic benchmark for evaluating temporal encoding and system adaptability.

Each digit in the N-MNIST dataset is represented as a stream of events triggered asynchronously in response to changes in pixel intensity, capturing the digit's structure as a sparse, time-resolved sequence of spikes. Specifically, each digit in the N-MNIST dataset comprises four timeframe images with 28 × 28 pixels, showing the appearance and disappearance of the digit (The left panel of Figs. 5b, 1 data mapping stage). Therefore, each pixel position contains four data points representing the evolution of brightness in the grayscale. For simplicity, the grayscale images were converted to binary images, and then each pixel position constitutes a four-element vector with 16 configurations, ranging from (0,0,0,0) to (1,1,1,1). The 0 and 1 data can be represented by 0 V/2 ms and 3 V/2 ms voltage pulses in this work, where the time allocation of each pulse of the given four-element vector represents the temporal correlations between the timeframe images. In contrast, the relationship between the four-element vectors represents the spatial correlations between the pixels in the given N-MNIST data, represented by 784 four-element vectors.

To implement the physical RC using PHT memristors, each of the 784 four-element vectors can be input to a PHT memristor using voltage pulses that follow the specific four vector elements, as shown in the right panel of Figs. 5b, 1, data mapping stage. Each PHT memristor then becomes to have a specific conductance depending on the four input voltage pulses, constituting the 1 × 784 reservoir state vector. The detailed configuration of this reservoir state vector is primarily determined by the relaxation properties of the PHT memristors, which are modulated by varying the t_b and V_b. The research goal of this work is to maximize the separability of the reservoir by finding the optimum t_b and V_b values, as discussed below.

In this work, only 10 PHT memristors were used owing to the practical limitations of fabricating the 784 PHT memristors. Therefore, the following sequential writing and reading processes were adopted to process the N-MNIST data. For a given N-MNIST image, '3' in Fig. 5b, the four-element vectors corresponding to pixels 1 to 78 were sequentially applied to memristor 1, and each vector was converted to the conductance of memristor 1, which is saved in the host computer. In this way, the first 78 components of the reservoir state vector are acquired. A similar procedure is repeated to acquire the second 78 components of the reservoir state vector by applying the second set of 78 four-element vectors to memristor 2. The rest components of the reservoir state vector are achieved similarly, except that the memristors 7 – 10 handled 79 vectors. This type of distributed conductance mapping was conducted across the 10 memristors to account for device-to-device variation in the practical hardware implementation.

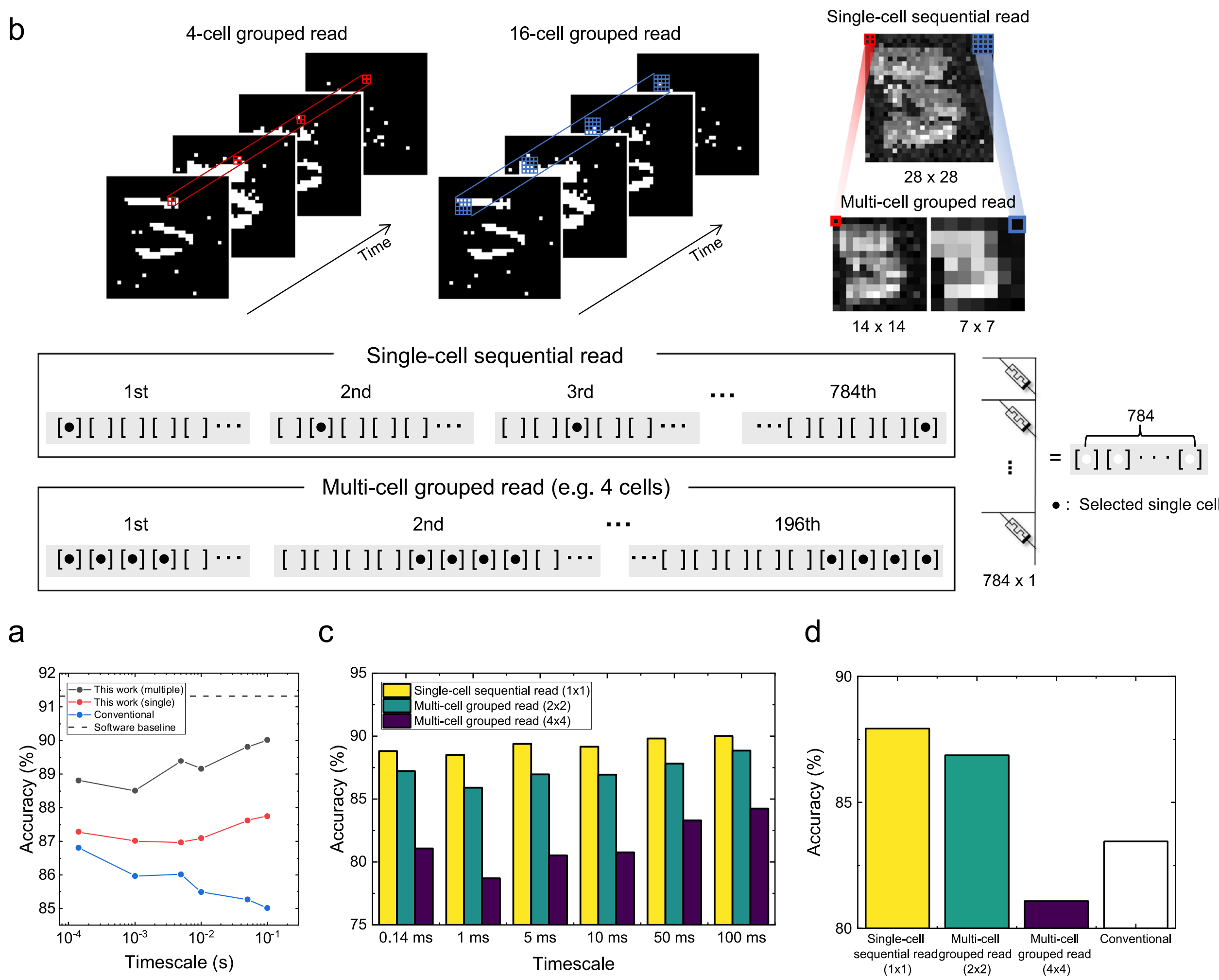

Fig. 6 | Grouped readout scheme and N-MNIST classification results. **a** Classification accuracies for different data timescales. While conventional RC accuracy declines as t_b exceeds the intrinsic τ, the V_b-modulated approach maintains high accuracy over three orders of magnitude in timescale. Using three V_b ('multiple') further increases the accuracy close to the software baseline. **b** Single-cell sequential readout and multi-cell grouped read. For multi-cell grouped read, multiple PHT cells are read in a group consisting of adjacent pixels in the original dataset. The two examples of 4-cell (colored red) and 16-cell (colored blue) grouped read are analogous to average pooling, which reduces read latency and readout layer size. **c** Classification accuracy from various multi-cell grouped read sizes. **d** Accuracy comparison in the preprocessed N-MNIST dataset with the randomized time intervals from 0.14 to 100 ms in Fig. 5c.

The second step of Fig. 5b displays the conductance mapping results of the N-MINST '3' digit obtained through the above-mentioned method, which are then flattened to input them into the 784 × 10 readout layer (Figs. 5b, 3 classification). During the entire process, converting each pixel image, represented by a four-element vector, to a conductance of the PHT memristor was performed by the hardware shown in Fig. 5a. At the same time, the remaining tasks were conducted through simulation using Python code. However, the crucial ingredient of the work is to enhance the separability of the input vectors by varying t_b and V_b. Figure 5c shows the measured conductance values after applying 15 patterns, except for the stream (0, 0, 0, 0), of 4-bit binary pulses. Here, the time interval between the four images (t_b) is varied from 0.14 to 100 ms to evaluate the tunability. Moreover, five V_b (0, 0.4, 0.8, 1.2, and 1.6 V) are used to manipulate the RC's timescale over a prolonged interval that exceeds the PHT's intrinsic timescale. The result shows an average from five independent measurements for each four-element vector with various t_b values. It shows that a prolonged data interval of over 10 ms degrades separability for the conventional RC with V_b set to 0 V. In contrast, increased V_b leads to higher separability even for long t_b, demonstrating its applicability to tunable data processes above the device's intrinsic τ.

However, the raw conductance map obtained through serial readout results in many reservoir states (784) and introduces latency due to the sequential single-cell access. In addition, the number of reservoir states increases linearly with the number of V_b employed. For example, the conductance map in the second panel of Fig. 5b shows the result for V_b of 0 V. Still, additional maps are generated as more V_bs are applied to expand the reservoir space, as further illustrated in Fig. 5c.

Figure 6a shows the classification accuracy for four cases: using V_b at 0 V (noted as 'Conventional'), a single V_b value (noted as 'Single'), three V_b values (0, 0.8, and 1.2 V) (noted as 'Multiple'), and the software baseline. For 'Single' and 'Multiple', the base voltages are chosen from the five V_b in Fig. 5c to achieve the highest classification accuracy. Specifically, V_b of 1.6 V was chosen for intervals between 0.14 and 10 ms, while 1.2 V was used for 50 and 100 ms in 'Single'. This outcome can be attributed to the separability trend shown in Fig. 5c, where higher base voltages generally enhance separability and consequently improve accuracy, except at longer timescales (t_b > 10 ms).

Comparing the 'Conventional' and 'Single' cases, which share the same readout layer size of 784 × 10, the conventional RC (blue line) shows a decline in accuracy as the data timescale exceeds the intrinsic τ. In contrast, the proposed method ('Single', red line) maintains stable accuracy across

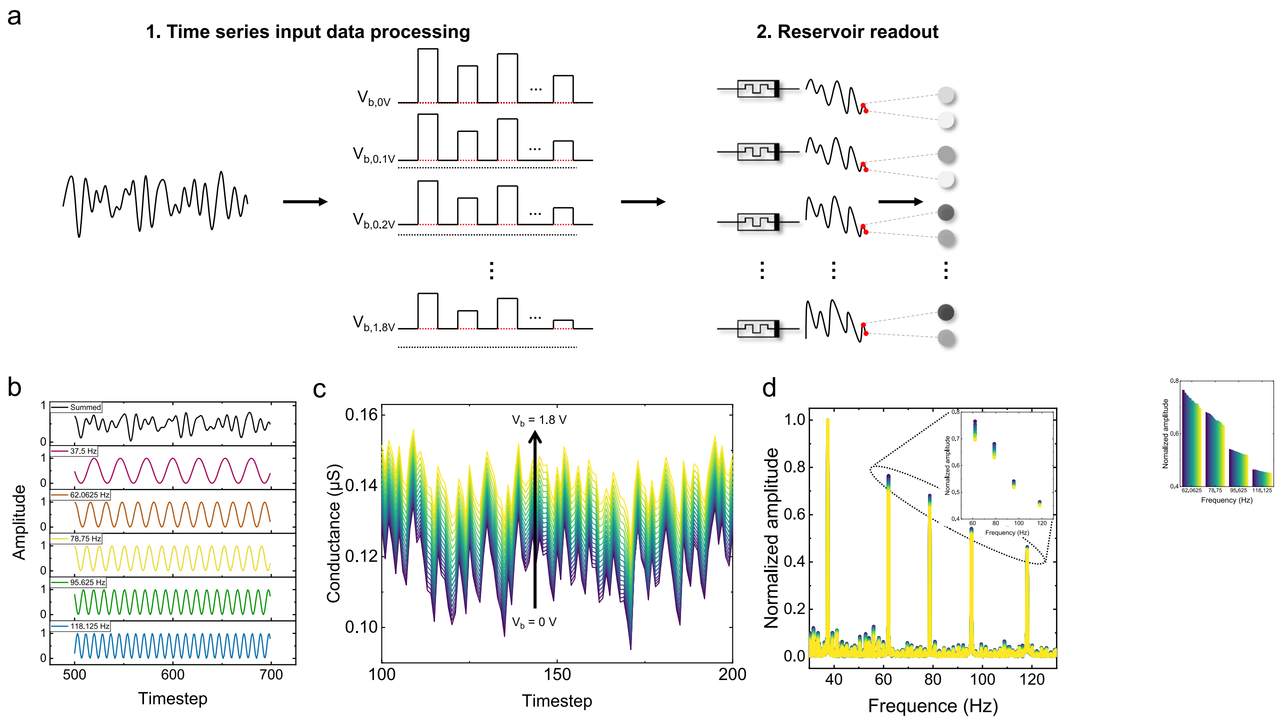

Fig. 7 | Analysis of temporal signal processing in the PHT under tunable V_b in the MSO task. a Schematic of the tunable V_b RC system. Analog time series inputs were applied to the PHT memristor with varied V_b (from 0 to 1.8 V), and the corresponding conductances were collected and used in the readout layer. The size of the readout layer varies depending on the number of virtual nodes in the MG time series prediction. **b** MSO time series data. The summed data (upper panel) comprises five superimposed sinusoidal functions with adjusted frequencies to align with the device's time scale. **c** Conductance responses of the PHT to MSO input signals with 19 V_b ranging from 0 V to 1.8 V for 100 timesteps. The legend of the V_b remains consistent across **c**–**e**. **d** FFT analysis of the conductance responses. The input frequency peak at 37.5 Hz was used for normalization. The inset shows the amplitude variation of input frequency peaks.

three orders of magnitude in timescale, suggesting that changing V_b effectively adapts the system to varying temporal scales. Moreover, the reservoir states can be expanded using multiple V_b to increase the classification accuracy. Specifically, it generates a conductance map for each V_b, which is concatenated and used as input to a readout layer with a size of 2352 × 10, three times larger than the original. The classification accuracy ('Multiple'), shown as the black line, is higher than that of 'Single', demonstrating the feasibility of state expansion. Moreover, it achieves a similar accuracy of 90.01% to the software baseline (91.32%), which used four raw timeframe images as input, resulting in a larger readout layer of 3136 × 10. It demonstrates that the proposed method achieves separability by enriching the reservoir states within a single device.

Although using three V_b increases the classification accuracy, it significantly increases the readout layer size. Figure 6b presents a multi-cell grouped readout method to address the readout inefficiency. In this method, each memristor corresponds to a specific pixel in the conductance map, and the readout layer size can be decreased by grouping memristors within a defined square region. This approach functions similarly to average pooling in conventional image processing. For instance, rather than reading all 784 conductance values sequentially, groups of PHTs can be read in parallel by applying read pulses simultaneously to a specific group of PHTs and summing the resulting current through a shared bit line. The group is decided on spatial proximity within the conductance map, and the group size determines the reduction factor. Specifically, red squares in the rightmost panel of Fig. 6b indicate 2 × 2 groups, where four PHTs are read together to decrease the layer size by a factor of 4. Blue squares indicate 4 × 4 groups, where sixteen PHTs are read together to decrease the size by a factor of 16. As a result, the proposed method decreases both the readout layer size and latency by summing the conductance values through grouped readout. This summing was conducted by the host computer using the estimated conductance values, not from the cross-line type hardware. Figure 6c shows the classification accuracies using the multi-cell read method to decrease the readout layer size. Across different timescales, the accuracy was maintained above 85% with 2 × 2 multi-cell read, using a decreased readout layer size of 588 × 10. This suggests that grouped readout effectively decreases the readout layer size with less than 2.5% degradation in accuracy.

Lastly, a modified N-MNIST dataset with random time intervals ranging from 0.14 to 100 ms was generated to evaluate the tunability of the RC and its adaptability to varying time scales (Fig. 6d). For each sample in the training and test datasets, the three intervals between the four timeframes are randomly chosen from the six t_b in Fig. 5c. This arrangement reflects a real-time data scenario where the interval between events varies unpredictably with asynchronous and temporally diverse input patterns. Three V_bs of 0, 0.8, and 1.2 V were used for classification, the same as 'Multiple' in Fig. 6c. The proposed method demonstrates higher accuracy (87.93%) than the conventional RC (83.45%). Moreover, the 2 × 2 multi-cell read also shows higher accuracy (86.87%) than the conventional RC, even with a decreased readout layer size (588 × 10 compared to 784 × 10). However, too much decrease in the reservoir output size (147) by adopting the 4 × 4 multi-cell read results in a degraded performance. The results indicate that the proposed method has higher tunability, allowing it to adapt to a wide range of temporal data.

MSO and bandwidth analysis

Like the N-MNIST dataset, time-series data in practical applications inherently contains multiple temporal information[34]. Hence, an RC system with high tunability across diverse timescales is required to extract features and improve computational efficiency. In the following sections, MSO and MG time series data are used to evaluate the RC's tunability and separability. Circuit simulation based on the measurement data was used in these sections (See the methods section for more details). Figure 7a describes the overall process of temporal data processing (MSO) and prediction (MG).

The continuous data is first encoded into analog voltage pulses. Then, the conductance change is read every timestep.

The MSO time series, defined as a sum of sinusoidal functions, is given by

$$u(t) = \sum_{i=1}^{5} \sin(2\pi\varphi_i t), \tag{1}$$

where φ_i represents the frequency of the i-th sinusoidal component. To fit the time scale of PHT, the φ_i values were scaled from the previous work[24] ($\varphi_1 = 37.5$, $\varphi_2 = 62.0625$, $\varphi_3 = 78.75$, $\varphi_4 = 95.625$, and $\varphi_5 = 118.125$). The uppermost panel of Fig. 7b shows the MSO time series data, where each frequency component (lower panels) is summed and normalized to make the amplitude unity. The time series consists of various frequencies, so it evaluates the system's tunability. While the MG time series exhibits chaotic behavior, the MSO time series can be accurately predicted given a sufficient readout layer size (Supplementary Fig. 5). Thus, the MSO time series is used to assess the tunability and separability of the reservoir without prediction. In contrast, MG prediction in the next section evaluates the RC's prediction performance.

Figure 7c shows the reservoir states of conductance from the MSO time series. First, the normalized time series was linearly transformed to input voltage pulses from 0 to 4.1 V with a 2 ms width. The interval between these input pulses, t_b, was set to 1 ms. Figure 7c shows 19 reservoir states by modulating V_b from 0.0 to 1.8 V in 0.1 V increments while applying 100 data points to the PHT serially. Each conductance value is read between the data pulses. Reservoir states with a higher V_b show significantly higher conductance, which coincides with the potentiation measurement in Fig. 4. Moreover, the reservoir state with V_b of 0.0 V changes more abruptly than those with higher V_b, because its relaxation timescale matches the device's intrinsic value, which is the shortest among the various reservoir states. In general, the reservoir states with lower V_b respond more significantly to input pulses, as they remain in the lower region of the possible conductance range, thereby further amplifying the differences between states. These findings suggest that separability can be feasibly obtained by adjusting V_b.

Then, the reservoir states were transformed into the frequency domain using the Fast Fourier Transform (FFT) for further analysis. Figure 7d displays the FFT spectra of the reservoir states, normalized to their maximum values at 37.5 Hz. Five distinct peaks in the frequency domain are observed at the same frequencies as the input signal, which are magnified in the inset. In a conventional RC system, when the input signal changes faster than the device's intrinsic τ, the device cannot track the input fluctuations, preventing effective encoding. Conversely, when the input timescale is much longer than τ, the signal is also not effectively encoded, as the reservoir cannot retain conductance changes over extended durations. Thus, conventional RC is unsuitable for real-time applications where the data timescale differs significantly from the device's intrinsic τ. The fixed data timescale also determines t_b, restricting input pulse width and interval manipulation to match the device timescale.

In contrast, Fig. 7d shows that the reservoir states generated under different V_b values exhibit varying response amplitudes depending on the input frequency. This indicates that the reservoir can effectively collect signals with t_b by varying V_b. Specifically, a short τ induced by low V_b allows the device to respond more strongly to high-frequency inputs. In contrast, a high V_b results in a longer τ, which allows the device to better respond to low-frequency inputs. Moreover, the reservoir states from multiple V_b in Fig. 7c can be used to increase the number of states and separability given a temporal dataset.

MG prediction

Finally, the MG time series[35] was used to demonstrate the suggested system's capabilities in predicting temporal data. The MG time series shows chaotic and non-linear behavior, requiring complex neural networks to predict[36]. The MG time series is generated from a nonlinear time-delayed differential equation, exhibiting a wide range of periodic and chaotic behaviors depending on the parameter values:

$$\frac{dx}{dt} = \beta \frac{x(t-\tau)}{1 + (x(t-\tau))^n} - \gamma x(t) \tag{2}$$

For this study, the parameters were set as $\beta = 0.2$, $\gamma = 0.1$, $\tau = 17$, n = 10, with an initial seed of $x(0) = 1.2$[15,20]. The overall process is similar to the MSO time series, except for the readout process. First, three PHTs with various V_b (0.3, 1.1, and 1.4 V) were assumed to expand the reservoir states. The normalized time series was transformed into the voltage pulses under the same conditions as MSO. Figure 8a illustrates the reservoir states with three V_b values obtained by applying 800 datapoints from the time series as voltage pulses. For one-step ahead prediction, the reservoir states were used to train a 150 × 1 readout layer, shown in Fig. 7a. Specifically, the reservoir receives the previous 50 data points leading up to the prediction point, with 50 virtual nodes from each of the three PHTs (or V_b) used to predict the next value. In this work, virtual nodes are created by reading the conductance between input data pulses, increasing the number of reservoir states derived from a temporal input[11]. Fig. 8b shows the prediction result obtained by using 800 steps as training data and 200 steps for prediction without a teaching signal to guide the prediction[15]. The normalized root mean squared error (NRMSE) for the prediction is 0.27, suggesting a feasible prediction using multiple V_b for reservoir state expansion.

Supplementary Fig. 6 shows the prediction results using a single V_b. Although all results utilized 150 virtual nodes to match the readout network size shown in Fig. 8b, the NRMSE varied for different values of V_b. Starting from the V_b of 0 V, the NRMSE decreases as the V_b increases, reaching the highest accuracy at 1.6 V, after which it increases again. The variation in NRMSE under different V_b values with a fixed reservoir size suggests that adjusting V_b allows τ to be tuned to the timescale most suitable for a given task, enabling optimal reservoir design.

Figure 8c compares the suggested RC with various neural networks, with the software-based approach noted as 'w/o RC' in the prediction accuracy. It shows the NRMSEs for the short-term prediction of 200 prediction steps. All the models' readout networks for RC consist of 150 input neurons without any hidden layers, including RC with multiple V_b for fair comparison. For instance, 'Conventional RC' (V_b of 0 V) and 'Single V_b' (V_b of 1.6 V) use 150 virtual nodes, whereas 'Double V_b' (V_b of 1.1 and 1.5 V) and 'Triple V_b' (V_b of 0.3, 1.1, and 1.4 V) use 75 and 50 virtual nodes, respectively. V_b was selected from the 19 V_b conditions (from 0 V to 1.8 V in 0.1 V increments) to maximize prediction accuracy. Linear, logistic, and tanh activation functions were used for neural networks without RC. In addition, both single-layer and two-layer readout networks are evaluated for the models without RC. Specifically, the leftmost 'Linear' model uses a 150 × 1 readout layer without activation. In contrast, the models with various activations use a 150 × N × 1 readout, where N is the number of hidden units indicated in parentheses.

The prediction error (0.27) is the lowest for 'Triple V_b', outperforming the software-based approach with various activation functions and hidden layers. The result also shows that the NRMSE is significantly lower in 'Single V_b' (0.51) than in 'Conventional RC' (0.78), indicating that modifying V_b improves prediction accuracy. Furthermore, using a larger number of V_b improves classification accuracy, even though they use the same readout network size of 150 × 1. This demonstrates that adjusting the number and combination of V_b captures time-series features optimally across low- and high-frequency components. Figure 8d compares the number of training parameters and NRMSE for the models discussed in Fig. 8c. The suggested RC exhibits high prediction accuracy with significantly fewer parameters than the software models. These results indicate that the proposed RC efficiently encodes temporal data through reservoir states with high separability and tunability, without requiring additional terminal or complex device structures.

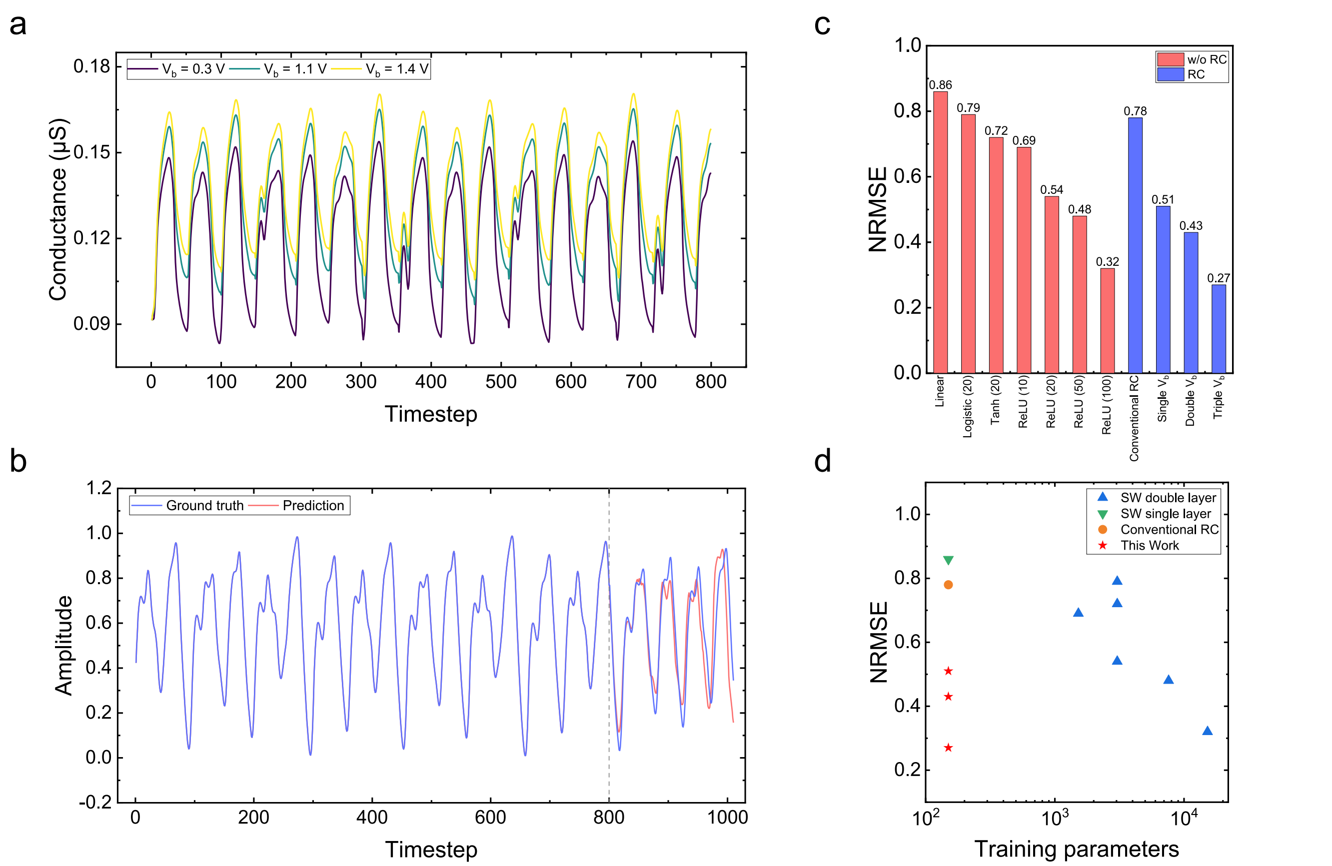

Fig. 8 | Performance evaluation of the proposed tunable V_b memristor RC system for MG time series prediction. **a** Conductance responses of the memristor under three different V_b (0.3, 1.1, and 1.4 V). **b** One-step-ahead prediction of the MG time series using the system with three V_b. The NRMSE is 0.27 with 800 and 200 timesteps for training and prediction, respectively. **c** Comparison of prediction accuracy between software-based models and memristor-based RC systems. **d** Prediction accuracy versus the number of parameters. It illustrates the efficiency of the tunable V_b memristor RC system compared to software-based models and conventional RC circuits, achieving higher accuracy with fewer parameters.

Discussion

This work demonstrated an electrically tunable physical RC system based on a two-terminal PHT memristor, achieving enhanced tunability and separability without complex device configurations or additional optical stimuli. By modulating the base voltage amplitude between data pulses, the device's intrinsic timescale was effectively extended by up to three orders of magnitude, ensuring that the full dynamic range of the device is harnessed even when processing extended temporal data. This tunability is critical for real-time applications where input timescales vary significantly.

Experimental validation using the N-MNIST dataset demonstrated that the proposed method maintains high classification accuracy across a broad range of data intervals and exhibits robust performance under variable temporal conditions. Furthermore, through analysis of MSO inputs and chaotic time-series prediction on the MG dataset, the proposed multiple V_b method produced diverse and rich reservoir states. These states enable accurate prediction with fewer parameters than conventional software-based models, suggesting the RC's separability and applicability to complex tasks.

This work measured the relaxation characteristic time as 1.2 ms, and the minimum pulse interval was set to 0.14 ms. However, previous work has shown that a device with the identical switching dielectric and switching mechanism ($W/HfO_2/TiN$) exhibits potentiation characteristics for pulses with a duration of ~100 ns[37]. Although the measurement setup in Fig. 5a and its parasitic elements limited the minimum timescale, an integrated system based on the scaled device would support datasets with timescales shorter or longer than the device's intrinsic limit. Moreover, negative base voltage will decrease τ below the device's physical limit as shown in Supplementary Fig. 8, expanding its applications to datasets with short timescales.

Overall, the proposed method enhanced tunability, scalability, and energy efficiency in physical RC systems by generating diverse, high-quality reservoir states with reduced device complexity. By effectively generating diverse, high-quality reservoir states with reduced complexity, the approach represents an advancement toward more efficient memristor-based RC implementations.

Methods

Device fabrication

The $Pt/HfO_2/TiN$ memristors were fabricated through a multi-step process. First, a 50 nm-thick TiN bottom electrode was deposited on a SiO_2/Si substrate via radio frequency (RF) sputtering (Sorona SRN 120). A titanium (Ti) target was used under a nitrogen (N_2) atmosphere at a pressure of 5 mTorr with an RF power of 100 W. The TiN layer was then patterned using maskless photolithography (Nano System Solutions Inc., DL-1000 HP) and defined through a conventional lift-off process. Next, a 4 nm-thick HfO_2 resistive switching layer was deposited via thermal atomic layer deposition (ALD) at a wafer temperature of 300 °C. $Hf(NC_2H_5CH_3)_4$ and ozone (O_3) at a concentration of 200 g/Nm^3 were used as the hafnium precursor and oxygen source, respectively. A 50 nm-thick Pt top electrode was then deposited via sputtering (SNTEK, CDS 5000) using power settings of 60 W. The deposition was performed at a working pressure of 1.6×10^{-2} Torr under an argon (Ar) flow of 30 sccm. The top electrode was patterned into orthogonal lines relative to the bottom electrode using conventional photolithography and lift-off processes. Post-fabrication structural and compositional analyses were conducted using transmission electron microscopy (TEM, JEOL, JEM-ARM200F) and energy-dispersive spectroscopy (EDS, JEOL, JEM-ARM200F).

Electrical measurements. A custom board, as shown in Fig. 5a, was used for the experimental demonstrations. It consists of a digital-to-analog converter, I/O interface, transimpedance amplifier, and analog-to-digital converter for physical RC evaluation. A microcontroller unit and Python code were used to control the integrated circuits. A pulse width of 2 ms was applied for both set and reset operations, enabling gradual voltage adjustments. During programming cycles, memristor conductance was monitored by applying a read voltage of 1.85 V, ensuring accurate state measurement without interfering with device operation.

Device modeling and simulation. The PHT's potentiation and relaxation behaviors were modeled for the physical RC simulations using voltage and time interval as parameters based on the measurement results shown in Figs. 2 and 3. Both logarithmic and exponential functions were used for fitting, according to various V_b. Then, the behavioral model was used to explain the conductance change for the temporal datasets. In addition to the behavioral model based on the fitting, C2C and D2D variations were considered in the simulation by using measured variations.

For the MG time series dataset, software-based simulations were conducted using the Python Scikit-learn library, employing both a linear regression model and a hidden-layer model as the readout layer. The accuracy of short-term prediction is evaluated using the NRMSE, which is calculated by dividing the root mean squared error by the prediction range. The dataset was first normalized to a range from 0 to 4.1 V and applied to the PHT device. Each pulse has a width of 2 ms with a 1 ms interval between consecutive pulses. V_b of 0.3, 1.1, and 1.4 V is maintained during the intervals (Fig. 8a-b). The preceding 50 timesteps are used as virtual nodes for each V_b condition. The measured read current was then fed into a single-layer readout network for prediction. To ensure a consistent reservoir state size of 150, the reservoir states were mapped to a weight matrix of dimensions 150×1 and optimized using Ridge regression.

Dataset preparation. The N-MNIST dataset was first preprocessed for use in the experimental demonstration. The event-based dataset was converted by setting the time interval (dt) to 20 ms, and four timeframes were extracted with a constant interval. Although the interval was specified in the data preprocessing, it was considered flexible, ranging from 0.14 to 100 ms, for the RC's tunability evaluation. Specifically, the initially modified dataset includes six time intervals (0.14, 1, 5, 10, 50, and 100 ms), with each N-MNIST data consisting of three time intervals between the four timeframes (Fig. 5b) set to a fixed time interval. In contrast, in Fig. 6d, each time interval was randomly selected. For example, a modified dataset might include four time frames of 0.14, 5, 1, and 5 ms, randomly assigned to all training and test images, respectively.

Data availability
The N-MNIST dataset is available at https://www.garrickorchard.com/datasets/n-mnist. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Code availability
The code of this study is available from the corresponding author upon reasonable request.

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Author contributions

S.H. Lee and S. Kim contributed equally to this work and designed the study concept and approach. S. Kim and K. Son fabricated the device. S.H. Lee, S. Kim, and J. Han performed electrical measurements. S.H. Lee and S. Kim designed and coded the experimental hardware system. S.H. Lee and S. Kim performed simulations. D.H. Shin, S.K. Shim, H. Jeong, J. Park, S.J. Shin, S. Ki,m and S.H. Lee analyzed the data. S.H. Lee, S. Kim, Y.H. Jang, and C.S. Hwang wrote the manuscript. Y.H. Jang and C.S. Hwang supervised this work.

Competing interests

The authors declare no competing interests.

Additional information

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